Health Risk Prediction

Under Distribution Shift

Robust Machine Learning Across Heterogeneous Patient Populations

CS4082 Machine Learning · Effat University · Spring 2026

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Problem, Motivation & Research Question

What is Distribution Shift?

Occurs when deployment data differs from training data.

- Different hospitals
- Different patient populations
- Different disease prevalence
- Different feature distributions

Why It Matters

A model may perform well during development but fail critically in real-world deployment.

Research Question

Can models trained on one cardiovascular dataset remain reliable under distribution shift, and which mitigation strategy improves performance the most?

We investigate 5 mitigation strategies across a full ML pipeline to answer this question.

Datasets Used

SOURCE DOMAIN

UCI Heart Disease

Patients:	**918**
Features:	**11 clinical**
Positive rate:	**55.3%**
Role:	**Train & Validate**

TARGET DOMAIN (OOD)

Framingham Heart Study

Subjects:	**4,240**
Features:	**15 epidemiological**
Positive rate:	**15.2%**
Role:	**OOD Evaluation Only**

Shared Predictors Used: Age · Sex · Cholesterol · Blood Pressure

Methodology Pipeline

1

Data Preparation

Clean, encode & align both datasets

2

EDA

Quantify distribution shift evidence

3

In-Domain Baselines

Train 7 models on UCI

4

Baseline OOD

Evaluate all models on Framingham

5

Class Weighting

Adjust loss for class imbalance

6

SMOTE

Synthetic oversampling in source domain

7

Hyper. Tuning

Grid search on UCI validation set

8

Threshold Tuning

Optimal threshold UCI val → Framingham

9

Pooled Training

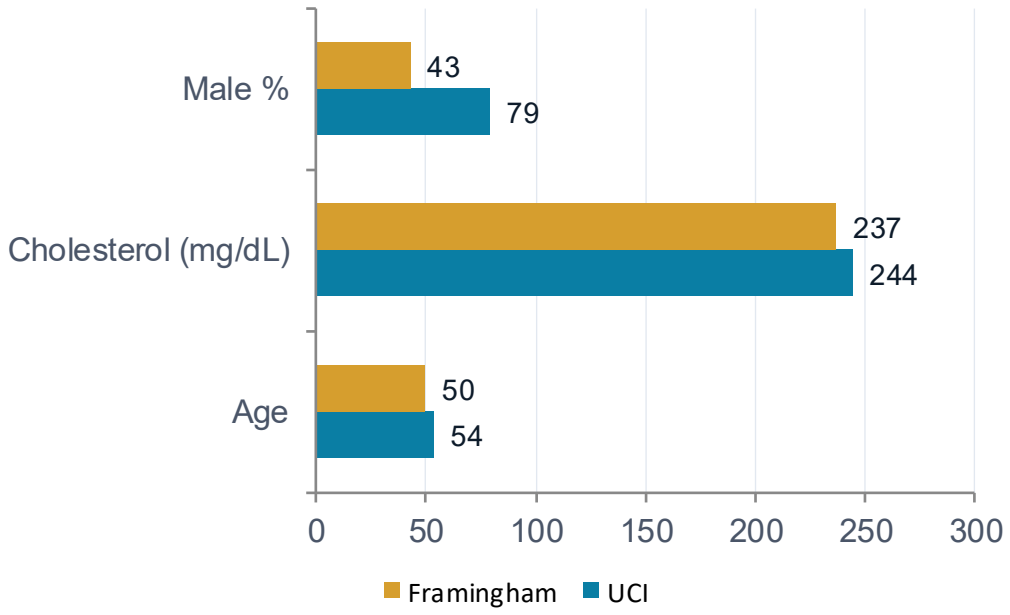
Train on UCI + 70% Framingham

10

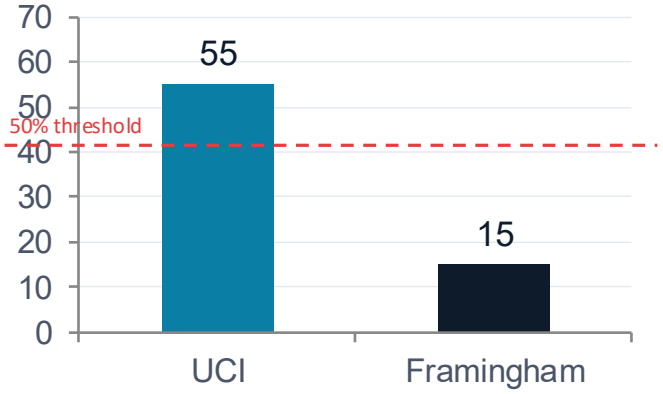
Final Comparison

Compare all strategies

EDA — Evidence of Distribution Shift



Shared Population Differences



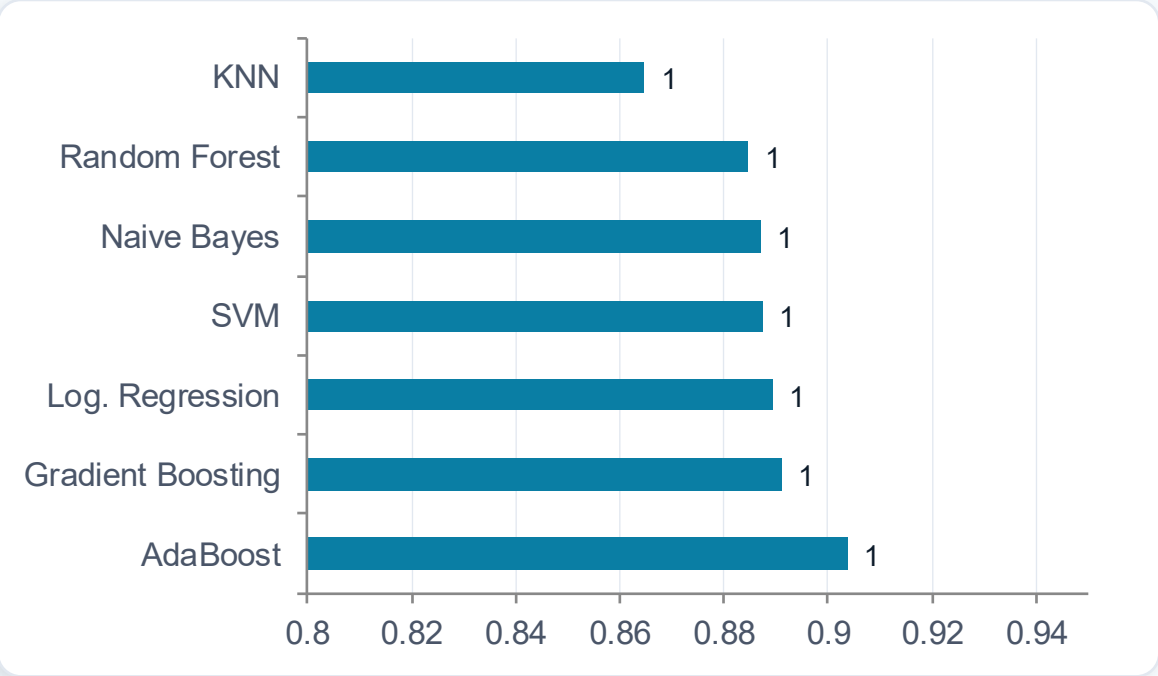
Key Findings

- Severe label shift: positive rate drops 3.6×
- Covariate shift confirmed (KS test $p < 1e-10$)
- Strong UCI predictors absent in Framingham

Distribution shift is severe and multi-dimensional — weaker OOD performance is expected.

In-Domain Baseline Results

Models trained and tested on the same distribution (UCI only).



0.9036

Best AUC — AdaBoost

~0.85

Average F1 Score

Key Insight: Strong performance when train/test distributions match — this is the in-domain ceiling.

Baseline Out-of-Domain Results

Train on UCI → Test on Framingham (never seen during training)

−0.22

AUC Drop (avg)

−0.51

F1 Drop (avg)

Key Insight: Major performance drop under distribution shift — mitigation is necessary.

Mitigation: Class Weighting & SMOTE

01 Class Weighting

Adjust the loss function to penalize minority class misclassification more heavily.

Best Model:	**SVM (balanced)**
Best AUC:	**0.6826**
Best F1:	**0.3611**
Recall:	**0.4876**

02 SMOTE

Synthetic Minority Oversampling — generates artificial minority samples in source domain.

Best Model:	**GB + SMOTE**
Best AUC:	**0.6877**
Best F1:	**0.3351**
Recall:	**0.4503**

Key Insight: AUC remained stable (~0.68) across balancing strategies — ranking ability is preserved despite calibration shift.

Mitigation: Hyperparameter & Threshold Tuning

03 Hyperparameter Tuning

Grid search on UCI validation set. Best configuration applied to Framingham.

Best Model:	**GB (tuned)**
Best AUC:	**0.6854**
Best F1:	**0.3491**
Recall:	**0.5606**

04 Threshold Tuning

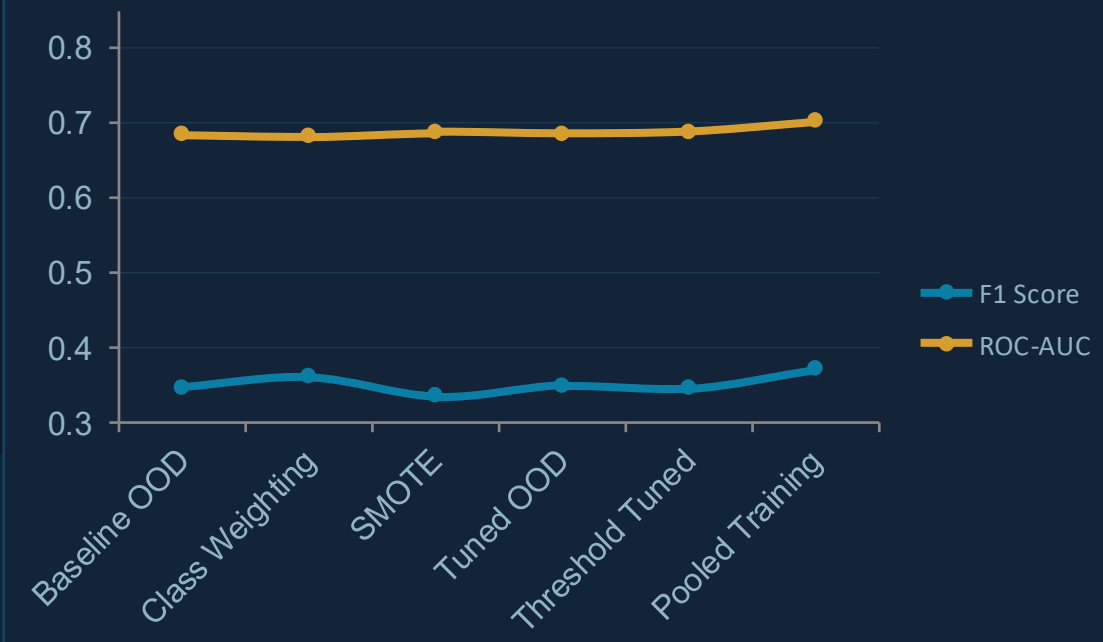
Optimal decision threshold from UCI val set applied to Framingham — no data leakage.

Model:	**Random Forest**
AUC:	**0.6882**
F1:	**0.3453**
Recall:	**0.7438 ↑**

Key Insight: Threshold tuning achieved the highest recall (0.74) — most effective when missing positives is the primary risk.

Best Final Results — Pooled Training

Train on: Full UCI + 70% Framingham → Test on: 30% Framingham holdout

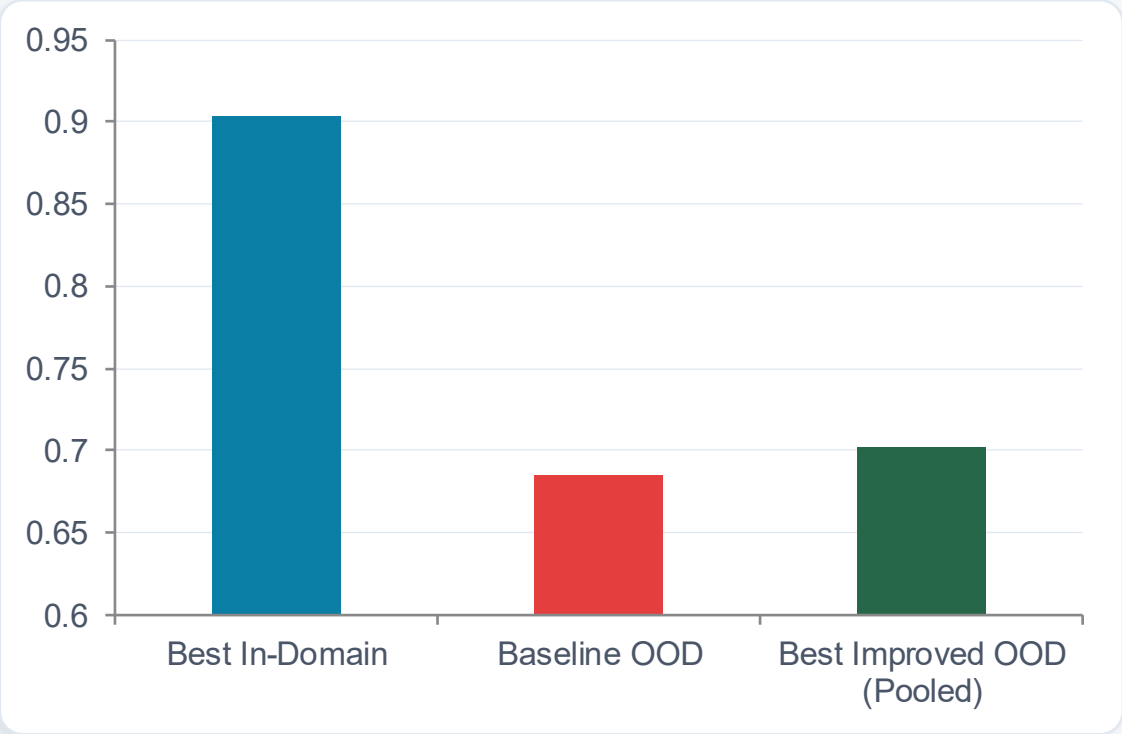


Model	ROC-AUC	F1
Logistic Regression	0.6925	0.2418
Gradient Boosting	0.6911	0.1908
Random Forest	0.6617	0.2382

Best AUC: 0.7125 (LR)

Precision improved — different recall tradeoff

In-Domain vs Best OOD Recovery



Scenario	ROC-AUC
Best In-Domain	**0.9036**
Baseline OOD	**0.6848**
Best Improved OOD	**0.6925**

Recovery: +0.0077 AUC

Gap remaining: 0.2111 AUC
AUC stable — calibration shift persists

Key Insight: AUC remained stable across all strategies (~0.68). Pooled training shifted the precision-recall tradeoff but did not close the domain gap.

Final Conclusions & Limitations

Conclusions

- Distribution shift significantly harms model reliability
- Class balancing gave small but consistent gains
- Hyperparameter tuning improved results further
- Pooled training achieved best final performance

Limitations

- Only 4 shared predictors — severely limits model expressiveness
- Different label definitions: exam-time disease vs 10-year CHD risk
- Some phases used different evaluation set sizes

Final Takeaway

More diverse training data improves robustness more than isolated model tuning.